

Deploying a Rapid Sensor-Fusion Data Collection Device to Automate Scrap Metal Sorting



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Limitations of Current Scrap Industry

The United States is Behind in the Scrap Industry

A strong domestic scrap industry is important for various reasons: security, economy, sustainability, and etc. It's even more important with the energy transition advancing, and more resources, such as metal scrap, being used to update infrastructure. Previously, the U.S. threw most of their scraps overseas to avoid the heavy labor and prejudice associated with recycling. Now the U.S. is severely behind in terms of throughput rates, range that can be recycled, and amount there is to be recycled. Current technology now relies on heavily experienced workers and X-ray fluorescence guns at the sorting phase, issues being workers are dwindling and XRF guns can't see past layers of paint, oxidation, and etc.



Figure 1: this is all copper scrap, separated by grades of copper, No1, No2

Why Copper?

Among the nonferrous (contains little to no iron) scrap metal industry, copper has become alarmingly crucial for the energy transition and is among the more difficult metals to identify when mixed with scrap with brass, solder, rust, etc. Copper also has strict requirements for manufacturing and the prices depend heavily on those grades, which scrapyards struggle with.

Objectives

Direction

To increase the rate of copper regeneration, our project focuses on automating sorting processes so that sorters can make quicker and more confident classifications of scrap metal. Our system plans to use **sensor fusion** to handle identification objectively and consistently, meaning facilities can expand their workforce without requiring years of training for workers [1][2]. To account for the shortcomings of XRF, we will also be implementing **Magnetic Induction Spectroscopy (MIS)** as a sensor that can probe beneath surfaces and coatings [3].

Steps to Get There

- In order for us to provide consistent and confident data rapidly, we need to train an ML model with data (LOTS of data).
- This means we need a system that can rapidly collect tons of data
And we need to collect data on not just various grades of copper but commonly confused metals, aka other nonferrous metals



Figure 2: Classifications of metals that we need to focus on and collect more data on, and are also important to scrap sorters

Converting Old Set Up to Field-Testable Device

Research and Planning

- Determining what samples are in scope and which aren't.
- Determining which sensors and methods that are worth testing within our scope.
- Designing for feasibility, useful data, and within the summer time frame.

Testing and Optimizing Systems

- Testing until results are repeatable, consistent, and accurate.
- Reducing scan time to increase scans per minute.

Data Collection

- Obtain individual sensor data on as many metal scraps possible in an actual junkyard, particularly copper and contaminated/ambiguous scrap.
- Keep a robust device

Photos From Testing

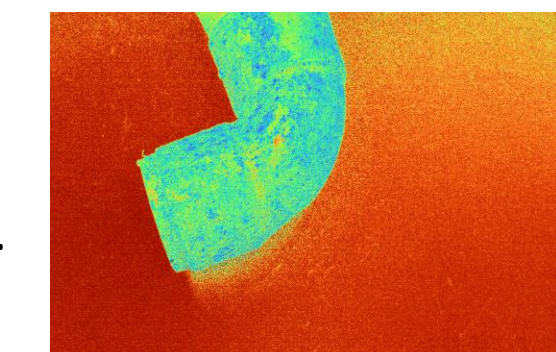


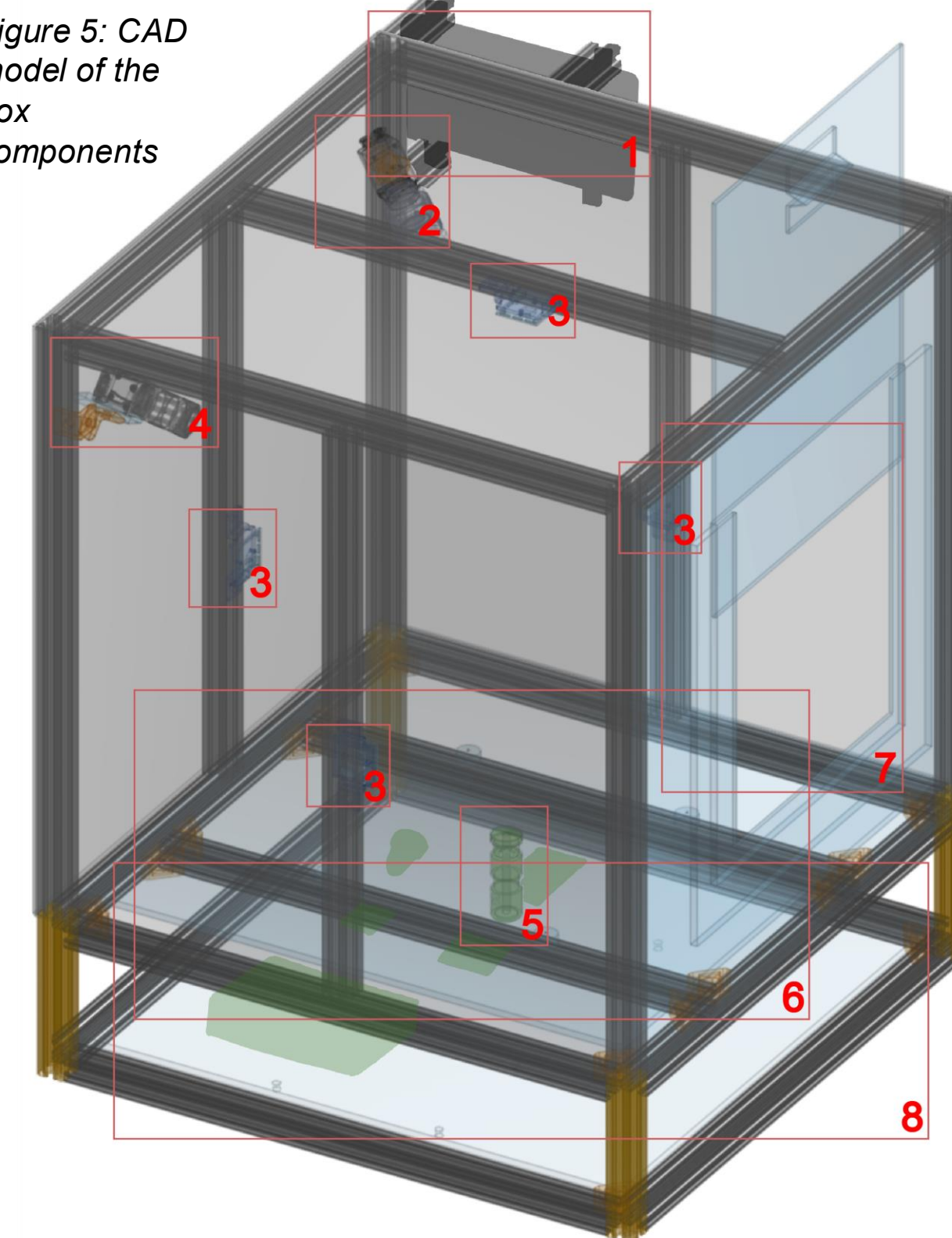
Figure 3: polarization camera degree of polarization view of heavily oxidized piece of copper plumbing



Figure 4: Basler camera view of same plumbing piece in Figure 2

My Focus and Contributions

Figure 5: CAD model of the box components



Research and Planning

- Researched the Manchester group extensively on MIS and applications used for metal sensing.
- Conducted 15+ interviews with scrapyards operators and metal recyclers to identify challenges in scrap identification and purchasing.

Testing and Optimizing Systems

- Optimized **MIS coil geometry**.
- Experimented with controlled environments.

Data Collection

- Obtained 600+ live metal scraps in an actual junkyard.
- Facilitated transportation and set up process of data collection with device.

Components

- Touchscreen connected to the RaspberryPi
- Basler Camera
- Polarization Camera
- RGB Cameras
- MIS Formers
- Base Plate
- Easy-access open door
- Electronics + MIS Compartment



Scan here for additional work and processes: our GitHub, User Manual, and other supporting works from this UROP

Results and Successes

Class Counters	
No. 1 Copper Solids:	62
No. 2 Copper Solids:	100
Red Brass:	102
Yellow Brass:	71
Aluminum:	64
Everything Else:	68

Figure 6: Tracker count at the end of on-site testing

Results

- Obtained data on ~600 distinct samples at Solomon Metals
- Increased brass dataset from the 5 samples in lab to 150+
- Learned more about scrap while implementing our device and communicating with scrap sorters in their day-to-day work

Successes

- Our device was able to handle ~6 hours of continuous scanning and most likely even more
- It proved robustness by enduring rough terrains during transport and easy cleanability while scanning

Future Applications

Post-Data Collection

With a trained ML, this data collection box should now be able to make inferences that has many new applications, some of those that have promising results would be:

- Shrinking volume of scanning area while maintaining or improving inference quality
- Fully automating the feeding process to demonstrate a proof of concept for scaling upwards at scrapyards
- Design a method to loosen the restraint on needing to place scrap directly above the MIS scanner

References

- [1] Diaz-Romero et al. [Classification of aluminum scrap by laser induced breakdown spectroscopy \(LIBS\) and RGB + D image fusion using deep learning approaches](#)
- [2] Gedam et al. [A Systematic Review: Development of AI Based Computer Vision Scrap Sorting System for Metal Scrap](#)
- [3] Williams et al., [Classifying Aluminum Scrap Metal Using Magnetic Induction Spectroscopy](#)

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